

REVERSE ENGINEERING

The Complete Professional Guide to Cybersecurity

Tools · Techniques · Methodology · Step-by-Step Workflows

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1. Introduction to Reverse Engineering

1.1 What Is Reverse Engineering?

Reverse engineering (RE) is the process of analyzing a system — software, hardware, firmware, or protocol — to understand its design, function, and operation without access to the original source code or blueprints. In cybersecurity, it is the foundational skill that enables security researchers to dissect malware, discover vulnerabilities, audit closed-source software, and develop defensive capabilities. Unlike traditional software development, which moves from specification to product, reverse engineering works backwards: from the compiled binary back to human-readable logic.

At its core, RE combines computer science, detective work, and creative thinking. A skilled reverse engineer must simultaneously understand low-level machine code, high-level program logic, operating system internals, and attacker psychology. This multi-disciplinary nature makes it one of the most challenging and most rewarding specializations in the entire field of cybersecurity.

1.2 History & Evolution

Reverse engineering predates digital computers — mechanical engineers have always disassembled competitors' products to understand their mechanisms. In computing, RE became significant in the 1980s when companies reverse engineered IBM's BIOS to create compatible PC clones, launching the personal computer revolution. The 1990s saw RE weaponized for both offense (early virus writers) and defense (the first antivirus researchers). The 2000s brought sophisticated nation-state malware: Stuxnet (2010) — a joint US/Israeli operation targeting Iranian nuclear centrifuges — was entirely uncovered through reverse engineering and remains the most studied piece of malware in history. Today, RE drives bug bounty programs, threat intelligence, and zero-day research at organizations like Google Project Zero, NSA, and every major cybersecurity vendor.

1.3 Why It Matters in Cybersecurity

- **Malware Analysis:** Understanding how malicious code operates enables defenders to build detection rules, patch victims, and attribute attacks.
- **Vulnerability Research:** Finding bugs in closed-source software before attackers do, enabling patches and responsible disclosure.
- **Penetration Testing:** Reversing client applications, APIs, and firmware to identify attack surface during authorized engagements.
- **Threat Intelligence:** Extracting indicators of compromise (IoCs), C2 infrastructure details, and actor TTPs from malware samples.
- **Protocol Reverse Engineering:** Reconstructing undocumented network protocols used by malware or proprietary devices.
- **Software Interoperability:** Creating compatible implementations when documentation is unavailable — the legal basis for tools like Wine.

2. Legal & Ethical Framework

■ **CRITICAL:** Before touching any tool or binary, understand that unauthorized reverse engineering of systems you do not own is a criminal offense in most jurisdictions. Always work within explicit legal authorization.

2.1 Key Laws & Regulations

Law / Regulation	Jurisdiction	Relevance to Reverse Engineering
Computer Fraud & Abuse Act (CFAA)	USA	Prohibits unauthorized access; RE on systems you don't own is illegal
Digital Millennium Copyright Act (DMCA) §1201	USA	Restricts circumventing DRM/copy protection; security research exemptions exist
Computer Misuse Act 1990	UK	Unauthorized access to computer material is a criminal offense
GDPR Article 32	EU	RE that exposes personal data requires careful handling and notification
Software Directive 2009/24/EC Art.6	EU	Permits RE for interoperability purposes only

2.2 Responsible Disclosure

When you discover a vulnerability through RE, responsible disclosure is the ethical and often legally protective path. The standard process: (1) Document the finding thoroughly. (2) Contact the vendor's security team privately (find security.txt or the security contact). (3) Give them a reasonable remediation window — typically 90 days, per Google Project Zero's standard. (4) If unresponsive, escalate to CERT/CC or publish a coordinated advisory. Never sell 0-days to brokers without understanding the legal and ethical consequences.

2.3 Bug Bounty Programs

Bug bounty platforms like HackerOne, Bugcrowd, and Intigriti provide legal frameworks for security research. Companies explicitly invite researchers to find vulnerabilities in their systems within defined scopes. Payouts range from \$100 for minor bugs to \$1,000,000+ for critical zero-days on platforms like Apple Security Research. Always read and respect the program scope — testing out-of-scope targets voids legal protections.

3. Foundational Knowledge

Reverse engineering without foundational knowledge is like trying to read a novel in an unknown language. You must be fluent in the following disciplines before attempting serious RE work.

3.1 Computer Architecture Essentials

You must deeply understand the **x86/x86-64** instruction set architecture (ISA) as it powers most desktop malware and CTF challenges. Also gain familiarity with **ARM** (mobile devices, embedded systems, Apple Silicon) and **MIPS/RISC-V** (routers, IoT).

Concept	Why It Matters for RE
Registers (EAX, RIP, ESP...)	Every instruction operates on registers; understanding their purpose lets you trace data flow
Stack & Calling Conventions	Function arguments, local variables, and return addresses all live on the stack — critical for exploit dev
Memory Segments (.text/.data/.bss)	Knowing where code, globals, and BSS live helps locate interesting data quickly
Endianness (little vs big)	Affects how multi-byte values are stored; critical when reading hex dumps
System Calls (syscalls)	Interface between user-space code and the OS kernel; malware heavily uses syscalls to evade user-mode hooks

3.2 Assembly Language Primer

You do not need to write assembly from scratch, but you must be able to read and understand it fluently. Compiled programs are displayed as assembly in disassemblers. Key instructions to master:

```
mov eax, 0x1337 ; Move value 0x1337 into register EAX
push ebp ; Save base pointer (function prologue)
call 0x401000 ; Call function at address 0x401000
cmp eax, 0 ; Compare EAX to zero (sets flags)
jz 0x40101A ; Jump if zero flag set (conditional branch)
lea ecx, [ebx+4] ; Load effective address – pointer arithmetic
xor eax, eax ; Fast way to zero EAX (common compiler idiom)
ret ; Return from function (pops RIP from stack)
```

3.3 File Formats & Binary Structure

Different platforms use different executable formats. A reverse engineer must recognize and parse each:

Format	Platform	Magic Bytes	Key Structures
PE (Portable Executable)	Windows (.exe/.dll)	4D 5A (MZ)	DOS header, PE header, section table, import/export tables
ELF (Executable & Linkable Format)	Linux/Android	7F 45 4C 46	ELF header, program/section headers, symbol table, dynamic section

Format	Platform	Magic Bytes	Key Structures
Mach-O	macOS/iOS	CE FA ED FE	Load commands, segments, fat binary headers (universal binary)
DEX (Dalvik Executable)	Android	64 65 78 0A	String/type/method pools, class definitions, bytecode

4. The Reverse Engineering Toolkit

Choosing the right tool for each task is half the battle. Below is a comprehensive breakdown of every essential tool, what it does, and when to use it.

4.1 Disassemblers & Decompilers

Tool	Type	Cost	Primary Use	Strengths
IDA Pro	Disassembler + Decompiler	\$1,499+	Industry-standard for malware analysis & vuln research	Best architecture support, Hex-Rays decompiler, mature plugin ecosystem (IDAPython)
Ghidra	Disassembler + Decompiler	FREE (NSA)	Full-featured alternative to IDA; great for beginners	Excellent decompiler, collaborative features, Java/Python scripting, runs on all OSES
Binary Ninja	Disassembler + Decompiler	\$299+	Automated analysis, API-first design, CI/CD integration	Python API, intermediate language (BNIL), headless mode, modern UI
Radare2 / Cutter	Framework + GUI	FREE	Command-line RE framework; Cutter is its Qt GUI frontend	Scripting (r2pipe), embedded systems, forensics, highly portable
RetDec	Decompiler	FREE	Automated decompilation to C, cloud or local API	Good for quick automated analysis, integrates with other tools

4.2 Debuggers

Tool	Platform	Best For	Key Features
x64dbg / x32dbg	Windows	Windows malware, crackmes, general RE	Open source, plugin system (ScyllaHide), excellent UI, active community
WinDbg (Preview)	Windows	Kernel debugging, crash dump analysis, driver RE	Microsoft official, kernel-mode support, time-travel debugging (TTD)
GDB + pwndbg / peda	Linux/macOS	Linux binaries, exploit development, CTF challenges	pwndbg adds heap visualization, ROP gadgets, PEDA adds exploit pattern tools
LLDB	macOS/iOS/Linux	Apple platform RE, Swift/ObjC binaries	Native Apple debugger, Python scripting, Xcode integration
OllyDbg	Windows (32-bit)	Legacy 32-bit Windows RE, learning debugger fundamentals	Classic tool, many tutorials available, good for learning

4.3 Dynamic Analysis & Sandboxes

Tool	What It Does	Output
Any.run	Interactive online sandbox; run malware and watch execution in real time	Behavioral reports, network PCAP, process tree, screenshots
Cuckoo Sandbox	Self-hosted automated malware analysis framework	API calls, registry changes, network traffic, dropped files
Process Monitor (ProcMon)	Real-time monitoring of filesystem, registry, and process activity on Windows	Detailed event log with filtering; essential for Windows malware analysis
Frida	Dynamic instrumentation toolkit; inject JavaScript into running processes	Function hooking, argument tracing, memory patching — works on all platforms
DynamoRIO / PIN	Binary instrumentation frameworks for instruction-level tracing	Execution traces, taint analysis, coverage data for fuzzing

4.4 Network Analysis Tools

Tool	Function	Key Capability
Wireshark	Packet capture & protocol analysis	Decode 1000+ protocols; follow TCP streams; export objects from HTTP/FTP
Burp Suite	Web application proxy & scanner	Intercept/modify HTTP/S; fuzz parameters; identify web vulns; essential for web app RE
Zeek (Bro)	Network traffic analysis framework	Generate structured logs from PCAP; excellent for C2 traffic analysis at scale
Scapy	Python packet manipulation library	Craft custom packets to test protocols; replay captures; RE network protocols
Fakenet-NG / INetSim	Network simulation for malware analysis	Simulate DNS, HTTP, SMTP etc. to capture malware network behavior without real connectivity

4.5 Hex Editors & Binary Analysis

Tool	Purpose
HxD (Windows)	Fast, free hex editor; inspect and patch binary files at the byte level
010 Editor	Template-based hex editor; binary templates for PE/ELF/ZIP/etc. let you parse structures visually
ImHex	Modern open-source hex editor with pattern language, data inspector, and entropy visualizer
Binwalk	Firmware analysis tool; scans binary files for embedded file signatures, compressed data, and filesystems
strings / floss	Extract printable strings from binaries; FLOSS also decodes obfuscated/stacked strings in malware

4.6 Specialized & Bonus Tools

- **angr**: Python framework for symbolic execution and automated binary analysis — find vulnerabilities without running the code.
- **QEMU**: Full-system emulator; run ARM/MIPS/PowerPC binaries on x86 — essential for IoT/firmware RE.
- **apktool / jadx**: Android RE: apktool decodes APKs and smali bytecode; jadx decompiles DEX to readable Java.
- **objection**: Runtime mobile exploration powered by Frida; bypasses jailbreak/root detection, traces calls.
- **Volatility 3**: Memory forensics framework; extract processes, network connections, registry, and malware from RAM dumps.
- **YARA**: Write patterns (rules) to classify and hunt for malware families across large file collections.
- **PE-bear / CFF Explorer**: Windows PE file analysis tools; inspect headers, sections, imports/exports, rich header.
- **VirusTotal**: Upload samples for multi-engine AV scanning plus behavioral reports; use for quick triage.

5. Static Analysis — Step-by-Step

Static analysis examines a binary without executing it. This is the safest first approach to any unknown sample. Follow this workflow:

5.1 Step 1 — Initial Reconnaissance (Triage)

Step 1: Identify the file type — Run `file malware.bin` (Linux) or check magic bytes in a hex editor. Never trust the file extension alone.

Step 2: Hash the sample — Compute MD5, SHA1, SHA256 with `sha256sum`. Search hashes on VirusTotal and MalwareBazaar for instant context.

Step 3: Check entropy — High entropy (>7.0) sections suggest packing/encryption. Use `binwalk -E` or IDA's entropy plugin. Packed samples need unpacking first.

Step 4: Extract strings — Run `floss malware.exe` — get URLs, IP addresses, registry keys, error messages, and decoded obfuscated strings in one shot.

Step 5: Scan for signatures — Run the binary through YARA rules (e.g., from the YARA-Rules GitHub repo) to instantly classify known malware families.

Step 6: Check PE/ELF headers — Open in PE-bear or CFF Explorer. Look for suspicious section names (.upx, .ndata), abnormal number of imports, or corrupted headers.

5.2 Step 2 — Disassembly Workflow in Ghidra

1. Launch Ghidra → New Project → Non-Shared → Import binary (drag & drop).
2. Double-click the imported file → click 'Yes' to auto-analyze. Use default analyzers + enable 'Decompiler Parameter ID'.
3. Wait for analysis. Open the Symbol Tree: Functions → look for 'main', 'WinMain', or DllMain as entry points.
4. In the Listing view, press 'G' to go to a specific address. Use 'Ctrl+Shift+F' to search for strings you found with FLOSS.
5. Right-click functions → Rename (press 'L') as you understand them: `sub_401000` → `decrypt_config`.
6. Use the Call Graph (Graph → Show Call Graph) to visualize which functions call which. Follow execution from entry point.
7. Use the Decompiler window (Window → Decompiler) for C-like pseudocode. This dramatically accelerates analysis.
8. Document findings in comments ('Semicolon' key inline, or `//` in decompiler view).

5.3 Step 3 — Import/Export & IAT Analysis

The Import Address Table (IAT) of a PE binary reveals a huge amount about its capabilities even before reading a single instruction:

API Call	Implied Capability
<code>CreateRemoteThread</code> , <code>VirtualAllocEx</code>	Process injection — code being injected into another process
<code>RegSetValueEx</code> , <code>RegOpenKeyEx</code>	Registry persistence — malware writing run keys to survive reboot

API Call	Implied Capability
InternetOpenUrl, HttpSendRequest	C2 communication — making HTTP requests to a command & control server
CryptEncrypt, CryptGenKey	Cryptographic operations — possibly ransomware encryption
GetAsyncKeyState, SetWindowsHookEx	Keylogging / input capture — credential theft behavior

6. Dynamic Analysis — Step-by-Step

Dynamic analysis runs the target binary in a controlled environment, observing its behavior in real time. It reveals runtime-only information: decrypted strings, network C2 addresses, dropped payloads, and anti-analysis checks.

6.1 Setting Up a Safe Lab Environment

■ **NEVER run malware on your host machine.** Use an isolated virtual machine. Here is the minimum recommended lab setup:

1. Install VMware Workstation Pro or VirtualBox on your analysis machine.
2. Create a Windows 10 VM (most malware targets Windows). Allocate 4GB RAM, 60GB disk.
3. Install analysis tools inside the VM: x64dbg, ProcMon, Wireshark, FakeNet-NG, PE-bear.
4. Take a 'clean' snapshot BEFORE any analysis — this is your restore point.
5. Configure the VM network to 'Host-Only' or run FakeNet-NG to simulate internet WITHOUT real connectivity.
6. Install FlareVM (Mandiant's free Windows RE toolkit) — it automates tool installation with one script.
7. For Linux analysis, use REMnux (reverse engineering Linux distro) as a companion VM.
8. Disable VMware/VirtualBox guest additions or hide VM artifacts — some malware detects virtualization and changes behavior.

6.2 Debugging a Binary with x64dbg — Full Workflow

Step 1 — Open the binary: Drag the sample onto x64dbg. It pauses at the system breakpoint (ntdll entry). Press F9 to run to the program's entry point.

Step 2 — Set breakpoints: Right-click → 'Follow in Disassembler' for interesting functions. Press F2 to toggle a software breakpoint. Set breakpoints on APIs identified in the IAT (e.g., right-click the import → 'Set Breakpoint').

Step 3 — Step through code: F7 = Step Into (follows calls), F8 = Step Over (executes calls without entering), F9 = Run until next breakpoint.

Step 4 — Inspect memory: In the dump pane, right-click register values (e.g., EAX pointing to a buffer) → 'Follow in Dump'. Watch encrypted data transform into plaintext.

Step 5 — Patch instructions: To bypass checks: double-click an instruction in the disassembly → modify it. Change conditional jumps (JZ/JNZ) to NOP or unconditional JMP to alter flow.

Step 6 — Dump unpacked code: If the binary decrypts a payload into memory, use Scylla (x64dbg plugin: Plugins → Scylla) to dump the unpacked PE from memory and fix the IAT.

6.3 API Monitoring with Frida

Frida lets you hook any function in a running process with JavaScript — no recompilation needed:

```
# Install Frida
pip install frida-tools

# List running processes
frida-ps -U # -U for USB device (mobile), omit for local

# Hook a function (JavaScript snippet)
frida -p -e "
Interceptor.attach(Module.getExportByName('kernel32.dll', 'CreateFileW'), {
```

```
onEnter: function(args) {  
console.log('CreateFile:', args[0].readUtf16String()); // Log filename  
}  
});  
"
```

6.4 Memory Forensics with Volatility 3

When you cannot run a debugger (e.g., analyzing a live incident or a full memory dump), Volatility 3 extracts forensic artifacts directly from RAM images:

```
python vol.py -f memory.dmp windows.pslist # List all running processes  
python vol.py -f memory.dmp windows.pstree # Process tree (shows parent-child relationships)  
python vol.py -f memory.dmp windows.netscan # Active network connections  
python vol.py -f memory.dmp windows.malfind # Find injected/suspicious memory regions  
python vol.py -f memory.dmp windows.dumpfiles --pid 1234# Dump files for a specific PID  
python vol.py -f memory.dmp windows.cmdline # Command-line args for each process
```

7. Malware Analysis Deep Dive

7.1 Malware Classification

Type	Primary Behavior	Key RE Focus
Ransomware	Encrypts files and demands ransom; typically uses AES+RSA hybrid encryption	Find encryption key generation, C2 for key escrow, file enumeration logic
RAT (Remote Access Trojan)	Provides attacker full control: keylogging, screenshots, file access, webcam	Reverse C2 protocol, extract hard-coded C2 address, decode config
Rootkit	Hides its presence by subverting OS structures (SSDT hooks, DKOM, bootkit)	Analyze kernel-mode components; use WinDbg kernel debugging; check DKOM artifacts
Botnet Agent	Receives commands from C2 for DDoS, spam, cryptomining, lateral movement	Decode C2 protocol (often HTTP/IRC/P2P), extract bot ID generation, decode commands
Stealer	Exfiltrates credentials, cookies, crypto wallets, browser data	Find targeted file paths, data exfiltration method (HTTP POST/SMTP), C2 endpoint

7.2 Unpacking & Deobfuscation

Most real-world malware is packed — the original code is compressed or encrypted, and a stub decrypts/decompresses it at runtime. Recognizing packing is essential:

- 1. Detect packing:** High entropy in sections + few meaningful imports (only LoadLibrary/GetProcAddress) = packed. Tools: Detect-It-Easy (DIE), PEiD, ExeinfoPE identify the specific packer.
- 2. Generic unpacking:** For UPX (most common packer): simply run ``upx -d malware.exe``. For custom packers: set a breakpoint on the OEP (Original Entry Point) — run until memory is decrypted, then dump with Scylla.
- 3. Find the OEP:** Common technique: set hardware breakpoint on execute at the stack pointer value at program start. The packer will jump to OEP after decryption. Alternatively, look for the 'tail jump' (JMP to a very different address after decryption loop).
- 4. String deobfuscation:** Use FLOSS for static decoding. For runtime decoding: set breakpoint after decryption function, inspect memory at the output buffer.
- 5. Script deobfuscation:** For PowerShell/JS/VBScript droppers: replace the execution call with Write-Output or console.log to dump decoded payload without running it.

7.3 C2 Infrastructure Mapping

Mapping the command-and-control infrastructure is a high-value output of malware RE. Once you extract a C2 IP or domain, you can build detection rules and pivot to discover the entire campaign infrastructure:

1. Extract C2 address from decoded config or strings — look for IP literals, DGA seeds, or encoded domains.
2. Use passive DNS (VirusTotal, Shodan, RiskIQ/PassiveTotal) to find all domains resolving to the same IP.
3. Run ``whois`` and certificate transparency (crt.sh) searches to find related infrastructure.
4. Analyze the C2 communication protocol: what HTTP user-agent does it use? What endpoints? What data format (JSON/custom binary)?
5. Write Suricata/Snort rules or YARA rules to detect C2 beacons on the network.

6. Submit findings to ThreatFox (abuse.ch) or ISACs to share IoCs with the broader community.

8. Vulnerability Research

8.1 Fuzzing Techniques

Fuzzing (fuzz testing) automatically generates malformed or random input to find crashes that indicate vulnerabilities. It is the most scalable method for finding bugs in closed-source software.

Fuzzer	Type	Best For
AFL++ (American Fuzzy Lop)	Coverage-guided grey-box fuzzer	Linux binaries with source code; state-of-the-art mutation strategies
WinAFL	Coverage-guided fuzzer (Windows)	Closed-source Windows applications using DynamoRIO for coverage
libFuzzer	In-process coverage-guided fuzzer	Library fuzzing; fast due to in-process execution; integrates with AddressSanitizer
Boofuzz	Protocol-aware network fuzzer	Network protocols and services; successor to Sulley

8.2 Identifying Memory Corruption Bugs

- **Buffer Overflow (Stack):** Local buffer written beyond its size, overwriting the return address. In 64-bit systems, DEP/ASLR/stack canaries are defenses; look for missing canaries (-fno-stack-protector).
- **Heap Overflow:** Heap chunk metadata or adjacent chunks corrupted by out-of-bounds write. Leads to heap metadata exploitation or UAF.
- **Use-After-Free (UAF):** Memory accessed after being freed — if attacker can control re-allocation, they control execution. Very common in browser exploits.
- **Format String:** User-controlled format string passed to printf family — allows arbitrary read/write. Look for ``printf(user_input)`` patterns.
- **Integer Overflow:** Arithmetic overflow in size calculations leads to undersized buffer allocation, then overflow. Common in file parsers.
- **Type Confusion:** Object treated as wrong type — common in C++ vtable manipulation and JavaScript JIT engines.

8.3 Writing a Proof-of-Concept Exploit

A PoC exploit demonstrates that a vulnerability is exploitable. For a classic stack buffer overflow on 64-bit Linux:

- 1. Find the offset:** Use ``pattern_create`` (pwndbg) or ``cyclic 200`` (pwntools) to generate a De Bruijn sequence. Run the target with this input and note the value of RIP at crash. Run ``cyclic -l`` to find exact offset.
- 2. Bypass DEP/NX:** Find ROP gadgets with ROPgadget or ropper: ``ROPgadget --binary vuln --rop``. Chain gadgets to call `mprotect()` or use `ret2libc` to call `system('/bin/sh')`.
- 3. Bypass ASLR:** Leak a pointer (format string bug, info leak) to defeat ASLR. Alternatively, use partial overwrites or brute-force if entropy is low (32-bit).
- 4. Write the PoC:** Use pwntools (Python) to automate: craft payload = padding + ROP chain. ``p = process('./vuln'); p.sendline(payload); p.interactive()``.

5. Document & Report: Capture the exploit with screenshots, write a clear bug report with CVSS score, reproduction steps, and impact. Submit via responsible disclosure.

9. Advanced Techniques

9.1 Anti-Reversing Bypass Techniques

Real-world malware and protected software employ anti-reversing tricks to hinder analysis. Recognizing and defeating these is a key professional skill:

Technique	How It Works	How to Bypass
IsDebuggerPresent	Checks PEB.BeingDebugged flag to detect debugger	ScyllaHide plugin in x64dbg patches this automatically; or NOP the check
Timing Checks (RDTSC)	Measures execution time; debuggers slow things down detectably	Patch RDTSC to return consistent values; use ScyllaHide's timing bypass
VM Detection (CPUID, registry)	Checks for VMware registry keys, CPUID hypervisor bit, artifacts like vmtools.exe	Remove VM artifacts from registry; patch CPUID; use bare-metal analysis for evasive samples
Code Obfuscation (junk code, opaque predicates)	Inserts meaningless instructions or always-true/false conditions to confuse RE tools	Use symbolic execution (angr) to prune dead branches; manually identify and NOP junk
Control Flow Flattening	Transforms structured code into a dispatcher switch — makes graph analysis hard	Use D810 (IDA plugin) or angr to de-flatten; trace execution to identify real paths

9.2 Firmware & Embedded Systems RE

IoT devices and network equipment run firmware — often compressed or encrypted filesystem images. This is one of the richest areas for vulnerability research:

1. Obtain firmware: download from vendor site, extract from device via UART/JTAG, or capture firmware update traffic.
2. Run binwalk on the firmware image: `binwalk -eM router_fw.bin`` — this recursively extracts and decompresses all embedded filesystems.
3. Navigate the extracted filesystem. Find the web server binary, config files, and any hardcoded credentials.
4. Use QEMU for emulation: `qemu-arm-static -L ./squashfs-root ./squashfs-root/usr/bin/httpd`` — run ARM binaries on x86.
5. Open interesting binaries in Ghidra (add the relevant CPU architecture plugin — Ghidra supports MIPS/ARM/PPC out of box).
6. Look for command injection in web CGI handlers, hardcoded credentials, and unauthenticated endpoints.

9.3 Android & iOS Reverse Engineering

Mobile RE requires platform-specific tools and knowledge of the respective runtimes (ART/Dalvik for Android, Objective-C/Swift runtime for iOS):

Task	Android Tool & Command	iOS Tool & Command
Decompile app	jadx-gui app.apk (Java decompile)	Hopper Disassembler / Ghidra
Decode resources	apktool d app.apk (smali + res)	class-dump / dsdump for ObjC headers
Runtime hooking	Frida / objection	Frida / Cycrypt / FLEX 3
Bypass root/jailbreak detect	objection --gadget 'app' explore → android root disable	Liberty Lite / objection jailbreak bypass
Traffic interception	Burp Suite + user-installed CA cert	Burp Suite + SSL Kill Switch 2

9.4 Symbolic Execution with angr

Symbolic execution treats program inputs as mathematical symbols, exploring all possible execution paths. angr can automatically find inputs that reach a target address (e.g., a 'success' path) — useful for crackmes, license checks, and CTF:

```
import angr
proj = angr.Project('./crackme', auto_load_libs=False)
state = proj.factory.entry_state()
simgr = proj.factory.simulation_manager(state)
# Find address of 'Correct!' message, avoid 'Wrong!'
simgr.explore(find=0x401200, avoid=0x401250)
if simgr.found:
    solution = simgr.found[0].posix.dumps(0) # stdin
    print('Password:', solution)
```

10. Building Your Pro Workflow & Career Path

10.1 CTF Challenges & Practice Labs

Capture The Flag (CTF) competitions are the best way to practice RE in a legal, gamified environment. Here are the top platforms to grind:

Platform	Focus	URL	Difficulty
pwn.college	Binary exploitation, RE fundamentals, structured curriculum	pwn.college	Beginner → Advanced
crackmes.one	Pure reverse engineering / crackmes in all languages	crackmes.one	All levels
HackTheBox	Full machines + dedicated RE & pwn challenges	hackthebox.com	Intermediate → Pro
PicoCTF	Educational CTF with RE, binary exploitation, forensics	picoctf.org	Beginner
MalwareBazaar	Real malware samples for analysis practice (free, legal sandbox)	bazaar.abuse.ch	Intermediate → Expert

10.2 Certifications & Learning Resources

- **GREM (GIAC Reverse Engineering Malware):** The gold standard RE certification. Covers static/dynamic analysis, unpacking, and malware behavior analysis. Highly respected by employers.
- **OSED (OffSec Exploit Developer):** Offensive Security's exploit development course and exam — x86 Windows exploitation, SEH, DEP/ASLR bypass, format strings.
- **CRTO / CRTE:** Certified Red Team Operator/Expert — includes RE for lateral movement and tool customization in realistic AD environments.
- **Malware Analysis Fundamentals (CISA):** Free online training from the US Cybersecurity agency — excellent for those entering the field.
- **'Practical Malware Analysis' (book):** Sikorski & Honig — THE textbook for malware RE. Read it cover-to-cover.
- **OpenSecurityTraining2 (ost2.fyi):** Free university-level RE courses: Architecture 1001, Malware Dynamic Analysis, Exploits 1003.
- **'The Shellcoder's Handbook':** Deep dive into vulnerability research, exploitation, and shellcode — essential for exploit developers.

10.3 Professional Career Roadmap

Reverse engineering skills open doors to some of the most impactful and well-compensated roles in cybersecurity:

Role	Primary RE Focus	Employers	Salary Range (USD)
Malware Analyst	Analyze malicious code for AV vendors, threat intel teams	CrowdStrike, Mandiant, ESET, Kaspersky, CISA	\$80k – \$160k

Role	Primary RE Focus	Employers	Salary Range (USD)
Vulnerability Researcher	Find and analyze zero-days in software and hardware	Google Project Zero, NSA, ZDI, Synack, MSRC	\$120k – \$300k+
Exploit Developer	Write weaponized code exploiting discovered vulnerabilities	Offensive security firms, government contractors, red teams	\$140k – \$400k+
Threat Intelligence Analyst	RE malware to attribute campaigns and build intel products	Recorded Future, Mandiant, Palo Alto Unit 42, MITRE	\$90k – \$180k
Reverse Engineer (Defense / Intel)	Nation-state malware analysis, firmware, hardware security	NSA/CISA/DARPA, GCHQ, BAE Systems, Booz Allen	\$100k – \$250k+

Suggested 6-Month Study Roadmap

Period	Theme	Activities
Months 1–2	Foundation	Learn x86-64 assembly (OpenSecurityTraining2 Arch1001). Read 'Practical Malware Analysis' Chapters 1-8. Set up FlareVM lab. Complete 20+ beginner crackmes.
Months 3–4	Core Skills	Master Ghidra and x64dbg (YouTube: OALabs, stacksmashing). Complete 'Practical Malware Analysis' fully. Analyze 10 real malware samples from MalwareBazaar. Join CTF team, do PicoCTF/HTB.
Months 5–6	Specialization	Pick a focus: malware analysis OR exploit dev. Start GREM or OSED prep. Write 2-3 public malware analysis reports. Contribute to open-source RE projects. Apply for junior positions or internships.

This guide is a living document. Reverse engineering is an ever-evolving discipline — stay current by following researchers on Twitter/X, reading Malware Traffic Analysis, the Mandiant blog, Project Zero blog, and attending conferences like DEF CON, Black Hat, and REcon. The best reverse engineers never stop learning.